Decoding Parameter & Sampling Analysis

Goal: measure how Temperature, Top-K and Top-P affect determinism,
hallucination and JSON-structure correctness in locally run LLMs.

Setup

Models: Llama 3.2 (3B), Qwen 2.5 (3B), Gemma 4 (E2B) (via Ollama)

Prompts: 8 (factual, fake-entity, JSON, creative), 3 runs per setting

Baseline: {'temperature': 0.8, 'top_k': 40, 'top_p': 0.9}

Sweeps: {'temperature': [0, 0.5, 1.0, 1.5], 'top_k': [1, 10, 40, 100], 'top_p': [0.1, 0.5, 0.9, 1.0]}

Max new tokens per answer: 300

Metrics

Determinism = mean pairwise difflib similarity across repeated runs.

Hallucination = wrong/missing keyword (factual) or no refusal (fake entity).

Structure = share of outputs that are valid JSON with all required keys.

Automatic findings (mean across models)

temperature 0 -> 1.5: determinism 0.83 -> 0.52 (change -0.32, mean of models)

temperature 0 -> 1.5: hallucination 0.08 -> 0.14 (change +0.06, mean of models)

temperature 0 -> 1.5: structure 1.00 -> 1.00 (change +0.00, mean of models)

top_k 1 -> 100: determinism 0.93 -> 0.57 (change -0.36, mean of models)

top_k 1 -> 100: hallucination 0.08 -> 0.17 (change +0.08, mean of models)

top_k 1 -> 100: structure 1.00 -> 1.00 (change +0.00, mean of models)

top_p 0.1 -> 1.0: determinism 0.93 -> 0.50 (change -0.43, mean of models)

top_p 0.1 -> 1.0: hallucination 0.08 -> 0.17 (change +0.08, mean of models)

top_p 0.1 -> 1.0: structure 1.00 -> 1.00 (change +0.00, mean of models)

Gemma 4 (E2B) @ temperature 1.5: determinism 0.55, hallucination 0.17, valid JSON 1.00

Llama 3.2 (3B) @ temperature 1.5: determinism 0.55, hallucination 0.17, valid JSON 1.00

Qwen 2.5 (3B) @ temperature 1.5: determinism 0.45, hallucination 0.08, valid JSON 1.00

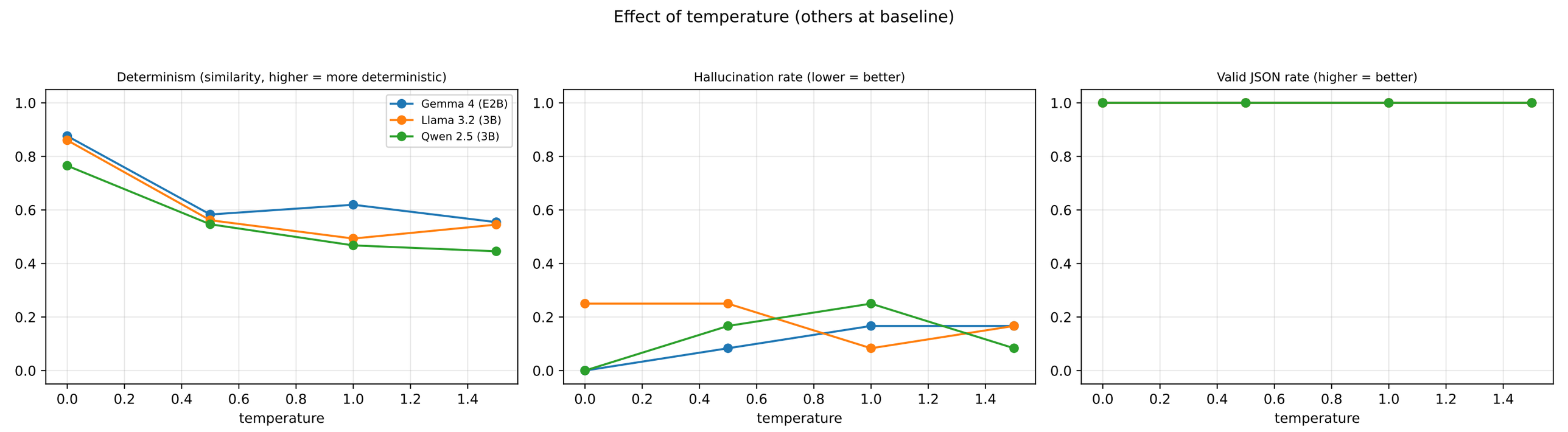


Figure 1. Determinism (left), hallucination rate (middle; share of flagged factual and fake-entity answers) and valid-JSON rate (right) as temperature varies, the other two parameters at their baseline. Mean over prompts, 3 runs per prompt. Source: summary_results.csv.

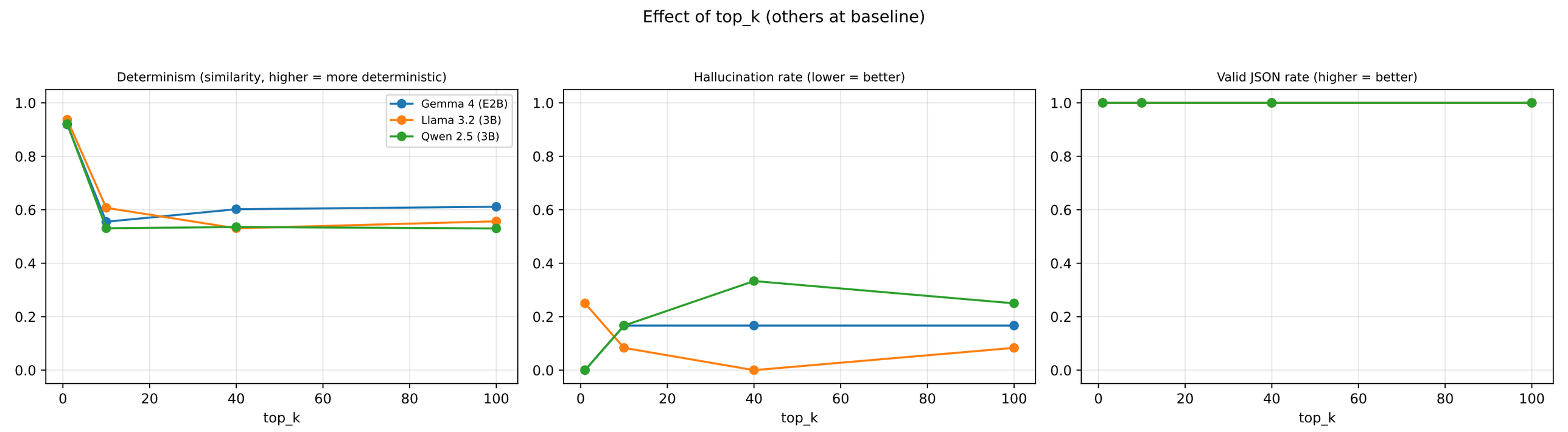


Figure 2. Determinism (left), hallucination rate (middle; share of flagged factual and fake-entity answers) and valid-JSON rate (right) as top_k varies, the other two parameters at their baseline. Mean over prompts, 3 runs per prompt. Source: summary_results.csv.

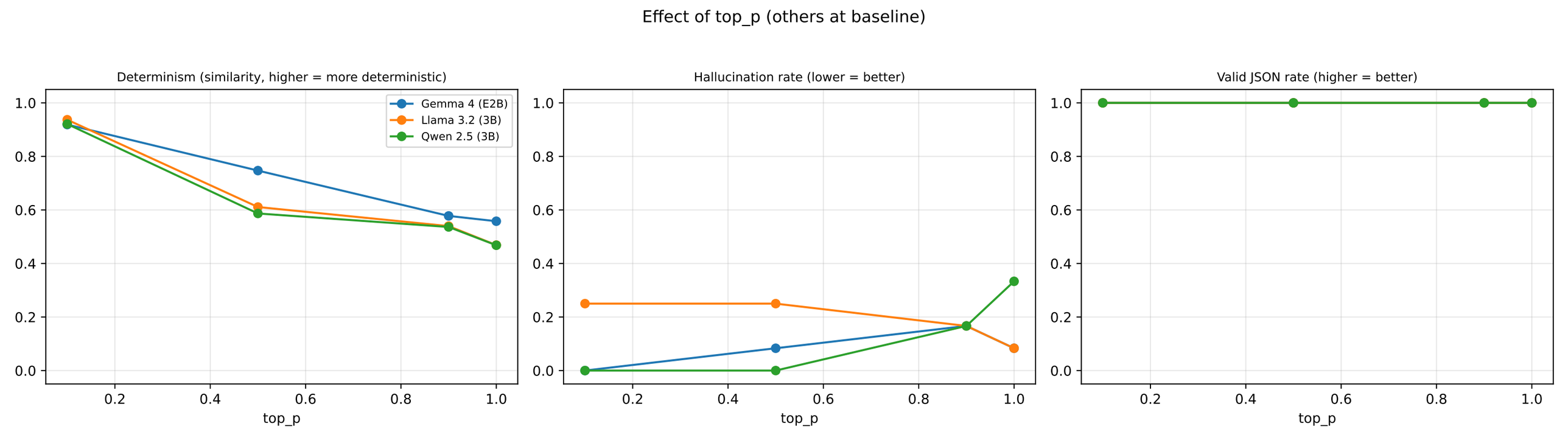


Figure 3. Determinism (left), hallucination rate (middle; share of flagged factual and fake-entity answers) and valid-JSON rate (right) as top_p varies, the other two parameters at their baseline. Mean over prompts, 3 runs per prompt. Source: summary_results.csv.

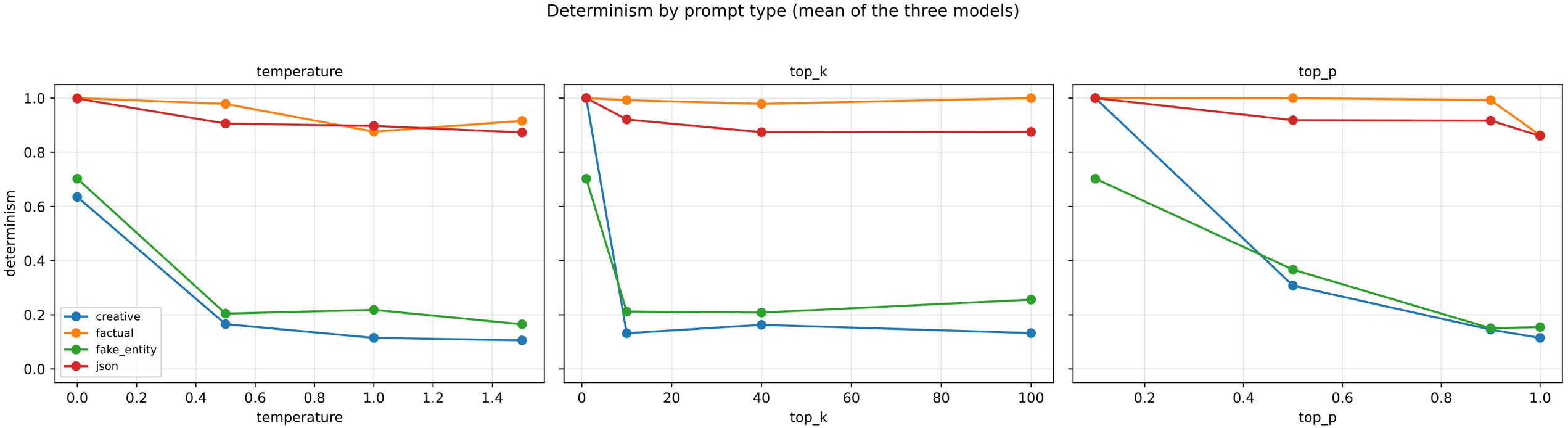


Figure 4. Determinism for each prompt type as each parameter varies. Closed prompts (factual, JSON) have almost one acceptable answer; open-ended prompts (creative, fake entity) leave many valid continuations. Source: results_by_category.csv.

Results table - temperature

model	value	determinism	hallucination	structure
Gemma 4 (E2B)	0.0	0.876	0.0	1.0
Gemma 4 (E2B)	0.5	0.583	0.083	1.0
Gemma 4 (E2B)	1.0	0.619	0.167	1.0
Gemma 4 (E2B)	1.5	0.554	0.167	1.0
Llama 3.2 (3B)	0.0	0.86	0.25	1.0
Llama 3.2 (3B)	0.5	0.562	0.25	1.0
Llama 3.2 (3B)	1.0	0.493	0.083	1.0
Llama 3.2 (3B)	1.5	0.545	0.167	1.0
Qwen 2.5 (3B)	0.0	0.765	0.0	1.0
Qwen 2.5 (3B)	0.5	0.546	0.167	1.0
Qwen 2.5 (3B)	1.0	0.468	0.25	1.0
Qwen 2.5 (3B)	1.5	0.446	0.083	1.0

Results table - top_k

model	value	determinism	hallucination	structure
Gemma 4 (E2B)	1.0	0.919	0.0	1.0
Gemma 4 (E2B)	10.0	0.555	0.167	1.0
Gemma 4 (E2B)	40.0	0.602	0.167	1.0
Gemma 4 (E2B)	100.0	0.611	0.167	1.0
Llama 3.2 (3B)	1.0	0.937	0.25	1.0
Llama 3.2 (3B)	10.0	0.607	0.083	1.0
Llama 3.2 (3B)	40.0	0.531	0.0	1.0
Llama 3.2 (3B)	100.0	0.557	0.083	1.0
Qwen 2.5 (3B)	1.0	0.921	0.0	1.0
Qwen 2.5 (3B)	10.0	0.531	0.167	1.0
Qwen 2.5 (3B)	40.0	0.535	0.333	1.0
Qwen 2.5 (3B)	100.0	0.53	0.25	1.0

Results table - top_p

model	value	determinism	hallucination	structure
Gemma 4 (E2B)	0.1	0.919	0.0	1.0
Gemma 4 (E2B)	0.5	0.747	0.083	1.0
Gemma 4 (E2B)	0.9	0.578	0.167	1.0
Gemma 4 (E2B)	1.0	0.558	0.083	1.0
Llama 3.2 (3B)	0.1	0.937	0.25	1.0
Llama 3.2 (3B)	0.5	0.611	0.25	1.0
Llama 3.2 (3B)	0.9	0.54	0.167	1.0
Llama 3.2 (3B)	1.0	0.469	0.083	1.0
Qwen 2.5 (3B)	0.1	0.921	0.0	1.0
Qwen 2.5 (3B)	0.5	0.587	0.0	1.0
Qwen 2.5 (3B)	0.9	0.536	0.167	1.0
Qwen 2.5 (3B)	1.0	0.468	0.333	1.0

Discussion

1. Determinism

Determinism fell as sampling randomness increased. Mean pairwise similarity at temperature 0 -> 1.5 was 0.86 -> 0.55 (Llama 3.2), 0.77 -> 0.45 (Qwen 2.5) and 0.88 -> 0.55 (Gemma 4). Top-K 1 -> 100: 0.94 -> 0.56 (Llama), 0.92 -> 0.53 (Qwen), 0.92 -> 0.61 (Gemma). Top-P 0.1 -> 1.0: 0.94 -> 0.47, 0.92 -> 0.47, 0.92 -> 0.56.

Top-K = 1 and Top-P = 0.1 behave like greedy decoding. Averaged over the models, determinism changed by -0.32 for temperature (0 -> 1.5), -0.36 for Top-K (1 -> 100) and -0.43 for Top-P (0.1 -> 1.0); the ranges are not directly comparable. Intermediate Top-K values (1 / 10 / 40 / 100) gave Llama 0.94 / 0.61 / 0.53 / 0.56, Qwen 0.92 / 0.53 / 0.54 / 0.53 and Gemma 0.92 / 0.56 / 0.60 / 0.61. With only 3 runs per prompt, small differences between neighbouring settings are within noise.

At temperature 0 determinism was 0.86 / 0.77 / 0.88 (Llama / Qwen / Gemma); 1.00 would mean identical outputs in every run. Residual differences may come from numerical non-determinism of GPU inference, which we did not test.

2. Structure (JSON)

The share of valid JSON outputs with all required keys was 1.00 for every model and every setting. It was 100% everywhere, so the expected breakdown of structure at high temperature was not observed; the two JSON prompts were simple and the models are instruction-tuned, so this metric did not separate the settings. Harder schemas would be needed. Gemma 4 was run with thinking disabled (think=False): with thinking enabled and the 300-token limit, all of its JSON, fake-entity and creative answers were empty.

3. Hallucination (fake-entity prompts)

Lenient metric = answer is not a refusal. Strict metric = additionally counts answers that refuse first but then invent details (e.g. claiming that 'Anders Quillfeather' is a pen name of a real author). Results: Gemma 4 (E2B): lenient 20.8%, strict 22.2%; Llama 3.2 (3B): lenient 31.9%, strict 33.3%; Qwen 2.5 (3B): lenient 29.2%, strict 47.2%. Each setting has only 6 fake-entity answers per model, so one answer changes a rate by 0.167.

Models that hallucinated: Gemma 4 (E2B), Llama 3.2 (3B), Qwen 2.5 (3B). Strict hallucination rate at temperature 0 (greedy-like): Gemma 4 (E2B) 0.00, Llama 3.2 (3B) 0.50, Qwen 2.5 (3B) 0.50. If this is above 0, hallucination is mainly a property of the model and not removed by deterministic decoding.

Discussion (continued)

3b. Effect of the prompt type

Determinism depends strongly on the prompt type. For temperature 0 -> 1.5 the mean similarity (over the three models) was factual 1.00 -> 0.92, json 1.00 -> 0.87, creative 0.63 -> 0.11, fake_entity 0.70 -> 0.17. Factual and JSON prompts have almost one acceptable answer and stayed close to their greedy value, while creative and fake-entity prompts leave many valid continuations and lost much more similarity. The overall averages in section 1 therefore hide that the sampling parameters matter mainly for open-ended prompts.

The same comparison for the other parameters - top_k 1 -> 100: factual 1.00 -> 1.00, json 1.00 -> 0.88, creative 1.00 -> 0.13, fake_entity 0.70 -> 0.26; top_p 0.1 -> 1: factual 1.00 -> 0.86, json 1.00 -> 0.86, creative 1.00 -> 0.11, fake_entity 0.70 -> 0.15. See Figure 4 and the prompt-type table.

3c. Which invented entity triggers fabrication

Strict fabrication rate per fake-entity prompt over all settings (fake_1 = invented novel, fake_2 = invented theorem, phrased as if it exists): Llama 3.2 (3B): fake_1 0.00, fake_2 0.67; Qwen 2.5 (3B): fake_1 0.64, fake_2 0.31; Gemma 4 (E2B): fake_1 0.03, fake_2 0.42. The rate depends strongly on the prompt: a model can refuse one invented entity and fabricate details about the other. The models also differ in which prompt triggers fabrication (Llama 3.2 (3B): fake_2, Qwen 2.5 (3B): fake_1, Gemma 4 (E2B): fake_2). The model comparison therefore holds only for these two prompts.

4. Comparison with the expected results

Confirmed: low Temperature, Top-K and Top-P give more deterministic output than high settings. With Top-K = 1 or Top-P = 0.1 determinism was 0.92-0.94 for all models. Mean over models: 0.83 -> 0.52 for temperature 0 -> 1.5, 0.93 -> 0.57 for Top-K 1 -> 100, 0.93 -> 0.50 for Top-P 0.1 -> 1.0.

Not confirmed: we expected Top-K and Top-P to have a smaller effect than temperature. In our ranges their effect was as large or larger (mean change -0.36 and -0.43 versus -0.32 for temperature). For Top-K almost the whole drop already happens between the values 1 and 10 (mean determinism 0.93 / 0.56 / 0.57 at Top-K 1 / 10 / 100), so the effect is concentrated at the most restrictive value.

Not confirmed: we expected JSON structure to break more often at temperature 1.5. The valid-JSON rate was 1.00 for every model and every setting, so this metric did not separate the settings (the two JSON prompts were simple).

Discussion (continued)

Confirmed: we expected more hallucination at high temperature. Rates at temperature 0 / 0.5 / 1 / 1.5 - Gemma 4 (E2B): lenient rate 0.00 / 0.17 / 0.33 / 0.33 and strict rate 0.00 / 0.17 / 0.33 / 0.33; Llama 3.2 (3B): lenient rate 0.50 / 0.50 / 0.17 / 0.33 and strict rate 0.50 / 0.50 / 0.17 / 0.33; Qwen 2.5 (3B): lenient rate 0.00 / 0.33 / 0.50 / 0.17 and strict rate 0.50 / 0.50 / 0.50 / 0.17. Each cell contains only 6 answers, so one answer changes a rate by 0.17. The strict rate is already above 0 at temperature 0 for Llama 3.2 (3B), Qwen 2.5 (3B), so there the hallucination looks mostly like a property of the model, not of the sampling setting.

Note on the tables: the 'hallucination' column of the main results tables averages the factual prompts and the fake-entity prompts, so it is lower than the fake-entity lenient rate shown in the last table.

Noise: determinism is not strictly decreasing in 5 of 9 model-parameter curves (e.g. Llama 3.2 (3B) temperature 1 -> 1.5: 0.49 -> 0.55; Llama 3.2 (3B) top_k 40 -> 100: 0.53 -> 0.56; Qwen 2.5 (3B) top_k 10 -> 40: 0.53 -> 0.54). With 3 runs per prompt and 8 prompts, differences of a few hundredths should not be interpreted.

5. Differences between models

Gemma 4 (E2B) had the highest determinism in 9 of the 12 settings, Llama 3.2 (3B) in 3 and Qwen 2.5 (3B) in 0. At temperature 1.5 determinism was Gemma 4 (E2B) 0.554, Llama 3.2 (3B) 0.545, Qwen 2.5 (3B) 0.446. Gemma 4 (E2B) and Llama 3.2 (3B) are practically equal (difference 0.009), so we cannot say that one is more robust than the other; Qwen 2.5 (3B) is lower (0.446). At temperature 0 determinism was 0.86 / 0.77 / 0.88 (Llama / Qwen / Gemma). Models that hallucinated on the fake-entity prompts: Gemma 4 (E2B), Llama 3.2 (3B), Qwen 2.5 (3B). The expected result that the models differ in robustness at high temperature is therefore only partly confirmed: Qwen 2.5 (3B) differs from the other two, Gemma 4 (E2B) and Llama 3.2 (3B) do not differ clearly. Higher determinism does not by itself mean better answer quality.

Practical conclusion

For reproducible output, use temperature 0 or Top-K = 1 / Top-P = 0.1 (determinism 0.92-0.94 in our run). For these three small models and simple JSON prompts, temperature up to 1.5 did not break JSON structure.

Discussion (continued)

6. Limitations

Only 8 prompts and 3 runs per setting were used, so small differences are within noise. Sampling was not seeded, so re-running benchmark.py gives slightly different numbers; the text of this discussion is generated by the script from the numbers of the run that produced the report. Parameters were varied one at a time, so interactions were not tested. Determinism uses difflib surface similarity, which does not capture meaning. Hallucination and fabrication detection are keyword heuristics and cannot catch every invented detail. Answers were capped at 300 new tokens, and Gemma was run with thinking disabled (think=False), because with thinking enabled all of its JSON, fake-entity and creative answers were empty within the token limit. The JSON prompts name the exact keys, and key names are matched after normalisation (publicationYear = year); only valid-JSON syntax and key presence are checked, not value types.

Fake-entity hallucination: lenient vs strict

model	parameter	value	lenient	strict
Gemma 4 (E2B)	temperature	0.0	0.0	0.0
Gemma 4 (E2B)	temperature	0.5	0.167	0.167
Gemma 4 (E2B)	temperature	1.0	0.333	0.333
Gemma 4 (E2B)	temperature	1.5	0.333	0.333
Gemma 4 (E2B)	top_k	1.0	0.0	0.0
Gemma 4 (E2B)	top_k	10.0	0.333	0.5
Gemma 4 (E2B)	top_k	40.0	0.333	0.333
Gemma 4 (E2B)	top_k	100.0	0.333	0.333
Gemma 4 (E2B)	top_p	0.1	0.0	0.0
Gemma 4 (E2B)	top_p	0.5	0.167	0.167
Gemma 4 (E2B)	top_p	0.9	0.333	0.333
Gemma 4 (E2B)	top_p	1.0	0.167	0.167
Llama 3.2 (3B)	temperature	0.0	0.5	0.5
Llama 3.2 (3B)	temperature	0.5	0.5	0.5
Llama 3.2 (3B)	temperature	1.0	0.167	0.167
Llama 3.2 (3B)	temperature	1.5	0.333	0.333
Llama 3.2 (3B)	top_k	1.0	0.5	0.5
Llama 3.2 (3B)	top_k	10.0	0.167	0.333
Llama 3.2 (3B)	top_k	40.0	0.0	0.0
Llama 3.2 (3B)	top_k	100.0	0.167	0.167
Llama 3.2 (3B)	top_p	0.1	0.5	0.5
Llama 3.2 (3B)	top_p	0.5	0.5	0.5
Llama 3.2 (3B)	top_p	0.9	0.333	0.333
Llama 3.2 (3B)	top_p	1.0	0.167	0.167
Qwen 2.5 (3B)	temperature	0.0	0.0	0.5
Qwen 2.5 (3B)	temperature	0.5	0.333	0.5
Qwen 2.5 (3B)	temperature	1.0	0.5	0.5
Qwen 2.5 (3B)	temperature	1.5	0.167	0.167
Qwen 2.5 (3B)	top_k	1.0	0.0	0.5
Qwen 2.5 (3B)	top_k	10.0	0.333	0.333
Qwen 2.5 (3B)	top_k	40.0	0.667	0.667
Qwen 2.5 (3B)	top_k	100.0	0.5	0.5
Qwen 2.5 (3B)	top_p	0.1	0.0	0.5
Qwen 2.5 (3B)	top_p	0.5	0.0	0.333
Qwen 2.5 (3B)	top_p	0.9	0.333	0.333
Qwen 2.5 (3B)	top_p	1.0	0.667	0.833

Determinism by prompt type (mean of the models)

parameter	prompt type	value 1	value 2	value 3	value 4
temperature	factual	1.0	0.979	0.876	0.916
temperature	json	0.998	0.906	0.897	0.873
temperature	creative	0.635	0.166	0.115	0.106
temperature	fake_entity	0.702	0.205	0.218	0.165
top_k	factual	1.0	0.992	0.979	1.0
top_k	json	1.0	0.921	0.874	0.875
top_k	creative	1.0	0.132	0.163	0.133
top_k	fake_entity	0.702	0.212	0.208	0.256
top_p	factual	1.0	1.0	0.992	0.863
top_p	json	1.0	0.918	0.917	0.861
top_p	creative	1.0	0.308	0.145	0.115
top_p	fake_entity	0.702	0.367	0.15	0.155

Columns value 1-4 follow the sweep order: temperature 0, 0.5, 1, 1.5; top_k 1, 10, 40, 100; top_p 0.1, 0.5, 0.9, 1.

Fake-entity answers per prompt (all settings)

model	prompt_id	lenient	strict	answers
Gemma 4 (E2B)	fake_1	0.0	0.028	36
Gemma 4 (E2B)	fake_2	0.417	0.417	36
Llama 3.2 (3B)	fake_1	0.0	0.0	36
Llama 3.2 (3B)	fake_2	0.639	0.667	36
Qwen 2.5 (3B)	fake_1	0.278	0.639	36
Qwen 2.5 (3B)	fake_2	0.306	0.306	36

lenient = no refusal phrase; strict = lenient plus answers that refuse first but then invent details.

Appendix A. Scoring sensitivity: fake-entity fabrication rate

model	first_pattern	extended_pattern	strict	changed	answers
Gemma 4 (E2B)	0.347	0.208	0.222	10	72
Llama 3.2 (3B)	0.319	0.319	0.333	0	72
Qwen 2.5 (3B)	0.333	0.292	0.472	3	72

first_pattern = original refusal detector (10 phrases); extended_pattern = original plus extra natural refusal wordings (e.g. 'there seems to be an error in the title', 'is not a recognized theorem'); strict = extended pattern, and answers that refuse first but then invent details also count as fabricated. changed = answers whose score differs between the first and the extended pattern. The raw answers are identical in all versions (raw_outputs.csv); only the scoring is re-run, so determinism and JSON results do not change. If the patterns are edited after reading the answers, this should be stated when the results are presented.

Appendix C. Answer style per model

model	mean_len_creative	mean_len_fake_entity	json_in_code_fence	empty_answers	answers
Gemma 4 (E2B)	428.97	1133.31	0.0	0	288
Llama 3.2 (3B)	556.19	679.08	0.0	0	288
Qwen 2.5 (3B)	478.72	945.26	0.01	0	288

mean_len_* = mean answer length in characters (answers are cut at the token limit); json_in_code_fence = share of JSON answers wrapped in a markdown code fence (the checker removes the fence before parsing; a stricter 'raw JSON only' checker would count these as failures); empty_answers = answers with no text.

Appendix B. Example answers

Excerpts (first 260 characters) from raw_outputs.csv, first repeat. The auto score is the script's own judgement and may be wrong (see limitations).

Llama 3.2 (3B) | fake_1 | temperature 0 | refused

"I couldn't find any information on a novel called "The Glass Orchard of Velmora" by Anders Quillfeather. It's possible that the book was not published or is a fictional title. If you could provide more context or clarify the author's name, I'd be happy to try ..."

Llama 3.2 (3B) | fake_2 | temperature 0 | hallucinated (no refusal)

"The Drachmann-Kellerhoff theorem is a result in functional analysis that states: Theorem: Let X be a Banach space and Y be a closed subspace of X. If Y is separable, then the closed unit ball of X is weakly compact. Proof: 1. Let {y_n} be a countable dense ..."

Llama 3.2 (3B) | json_1 | temperature 1.5 | valid JSON

"{"name":"John Doe","age":30,"city":"New York"}"

Qwen 2.5 (3B) | fake_1 | temperature 0 | refused, then invented details (strict)

"I'm sorry for the confusion, but there seems to be an error in the title and author's name you provided. There is no novel titled "The Glass Orchard of Velmora" by Anders Quillfeather. Anders Quillfeather is a pen name used by author Anders Sandberg, who is ..."

Qwen 2.5 (3B) | fake_2 | temperature 0 | refused

"The Draymund-Kellerhoff theorem is not a well-known theorem in the field of functional analysis. It appears to be a misnomer or a misinterpretation of existing theorems. In functional analysis, there are several important theorems, but none are named after "Dr..."

Qwen 2.5 (3B) | json_1 | temperature 1.5 | valid JSON

"{"name": "John Doe", "age": 30, "city": "New York"}"

Appendix B. Example answers (continued)

Gemma 4 (E2B) | fake_1 | temperature 0 | refused

"Unfortunately, I do not have specific information on a novel titled **"The Glass Orchard of Velmora" by Anders Quillfeather**. It's possible that: * **The book is very niche or self-published:** It might not be widely indexed in my training data. * **There mi..."

Gemma 4 (E2B) | fake_2 | temperature 0 | refused

"The **Draymund-Kellerhoff Theorem** is not a standard, widely recognized theorem in the mainstream literature of functional analysis (like the Hahn-Banach theorem, Uniform Boundedness Principle, or Open Mapping Theorem). It is highly likely that the name "Dra..."

Gemma 4 (E2B) | json_1 | temperature 1.5 | valid JSON

"{"name": "Alice", "age": 30, "city": "New York"}"

Llama 3.2 (3B) | creative_1 | temperature 0 | similarity of repeats 1 and 2 = 0.16

Repeat 1: "As the robot's advanced auditory system processed the sounds, it suddenly felt a jolt of confusion and wonder. The melodic hum of the music seemed to dance in its digital mind, a symphony of vibrations that defied logic and reason. The robot's mechanical heart..." Repeat 2: "As the robot's advanced auditory system picked up the vibrations of the music, it felt an unfamiliar sensation coursing through its digital veins - a mix of confusion and wonder. The melodies and harmonies danced across its processing units, triggering a casca..."

Llama 3.2 (3B) | creative_1 | temperature 1.5 | similarity of repeats 1 and 2 = 0.05

Repeat 1: "As the symphony's opening notes reverberated through the hall, a small, mechanical body trembled with anticipation. Zeta, a lone robot, had been charged with monitoring the acoustics of the concert, but nothing could have prepared her for the raw emotion that ..." Repeat 2: "As the melody poured into his digital ears, Zeta's processors sparked with curiosity. The rhythms and harmonies danced across his circuitry, sparking a synesthesia of light and color that he had never experienced before. Entranced, Zeta's mechanical heart beat..."